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1 September 1977

MILITARY STANDARD

DUMMY LOADS, ELECTRICAL, WAVEGUIDE, COAXIAL, AND STRIPLINE, SELECTION OF

FSC 5985

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DEPARTMENT OF DEFENSE
Washington, DC 20301

Dummy Loads, Electrical, Waveguide, Coaxial, and Stripline, Selection of.

MIL-STD-1637A

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ELEX 5043, Washington, DC 20360, by using the self-addressed Standardization *Document Improvement Proposal (DD Form 1426)* appearing at the end of this document or by letter.

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1. SCOPE

1.1 Scope. This standard provides standard waveguide, coaxial, and stripline dummy loads considered by the Department of Defense as standard for use in military equipment and applications.

1.2 Purpose. The purpose of this standard is to:

a. Provide the equipment designer with a list of dummy loads considered standard for use in military applications.
b. Restrict and minimize the variety of dummy loads for use in military applications in order to provide effective logistic support of equipment.
c. Establish criteria pertinent to choice and application of dummy loads for use in military equipment.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-D-3954 - Dummy Loads, Electrical, Waveguide, General Specification for.

MIL-D-39030 - Dummy Load, Electrical, Coaxial, General Specification for.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. DEFINITIONS

3.1 The terms used in this standard are those commonly encountered in microwave engineering practice.

4. GENERAL REQUIREMENTS

4.1 Selection of dummy loads. Dummy loads to be used in military applications shall be selected from those listed in tables I, II, and III.

4.2 Criteria for inclusion. The criteria for the selection of dummy load types for inclusion in this standard are:

a. The dummy loads shall be considered by representatives of the military departments the best available type for current application.
b. Availability of the dummy loads shall be reasonably certain.
c. The dummy loads shall have an approved military specification.

4.3 Electrical and physical tolerances. Dummy loads used in military applications shall be representative of manufactured lots possessing acceptable material and physical and electrical characteristics and shall in no manner degrade the operational characteristics of the equipment in which used.

5. DETAILED REQUIREMENTS

5.1 The detailed requirements for dummy loads listed in this standard are covered by the applicable MIL-D-3954 and MIL-D-39030 specification sheets.

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TABLE I. MIL-D-3954 waveguide dash numbers and characteristics.

Part number	Dash	Frequency range (GHz) Min	Frequency range (GHz) Max	VSWR (max)	Power Average	Power Peak	Pressure	Flange equivalent to	AN nomenclature	Flow rate	Material
					W	kW	psig			gpm	
CLASS I - DRY FINNED WAVEGUIDE LOADS											
M3954/15	01	1.12	1.70	1.1:1	8000	2200	23	M3922/58-008 (UG-418B/U)	DA-147/U	---	AL
M3954/18	01	2.60	3.95		4500	3200	30	M3922/56-002 (UG-584/U)	DA-145/U	---	
M3954/17	01	3.95	5.85		2000	1000	17.2	M3922/57-001 (UG-407/U)	DA-149/U	---	
M3954/16	01	5.85	8.20		1000	710	30	M3922/55-002 (UG-441/U)	DA-144/U	---	
M3954/11	01	7.05	10.00		600	460		M3922/53-004 (UG-138/U)	DA-148	---	
M3954/19	01	8.20	12.40		500	290		M3922/53-003 (UG-135/U)	DA-146/U	---	
M3954/13	05	12.40	18.00		200	10		M3922/53-006 (UG-1665/U)	---	---	
M3954/13	01	12.40	18.00	1.15:1	250	160		M3922/53-006 (UG-1665/U)	DA-159/U	---	
M3954/14	01	18.00	26.50		150	80	21.3	M3922/54-002 (UG-597/U)	DA-160/U	---	
M3954/12	01	26.50	40.00		75	100	30	M3922/54-003 (UG-599/U)	DA-158/U	---	
CLASS II - LIQUID COOLED WAVEGUIDE LOADS											
M3954/15	02	1.12	1.70	1.1:1	15000	17200	30	M3922/58-008 (UG-418B/U)	---	5	AL
	03							M3922/58-007 (UG-417B/U)	---		COP
	04							M3922/58-007 (UG-417B/U)	---		CRES
M3954/18	02	2.60	3.95		7500	3200		M3922/56-002 (UG-584/U)	---	2.25	AL
	03							M3922/56-001 (UG-53/U)	---		COP
	04							M3922/56-001 (UG-53/U)	---		CRES
M3954/17	02	3.95	5.85		4000	1300		M3922/57-001 (UG-407/U)	---	1.25	AL
	03							M3922/57-002 (UG-149/U)	---		COP
	04							M3922/57-002 (UG-149/U)	---		CRES
M3954/20	01	5.40	5.90	1.2:1	4500	3000		M3922/57-001 (UG-407/U)	---	1.3	AL
M3954/16	02	5.85	8.20	1.1:1	2000	710		M3922/55-002 (UG-441/U)	---	0.667	AL
	03							M3922/55-001 (UG-344/U)	---		COP
	04							M3922/55-001 (UG-344/U)	---		CRES
M3954/11	02	7.05	10.00		1500	460		M3922/53-004 (UG-138/U)	---	0.5	AL
	03							M3922/53-002 (UG-51/U)	---		COP
	04							M3922/53-002 (UG-51/U)	---		CRES

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TABLE I. MIL-D-3954 waveguide dash numbers and characteristics - Continued.

Part number	Dash	Frequency range (GHz) Min	Frequency range (GHz) Max	VSWR (max)	Power Average	Power Peak	Pressure	Flange equivalent to	AN nomenclature	Flow rate	Material
					W	kW	psig			gpm	
CLASS II - LIQUID COOLED WAVEGUIDE LOADS											
M3954/19	02	8.20	12.40	1.1:1	400	290	30	M3922/53-001 (UG-39/U)	---	---	COP
	03				1000			M3922/53-003 (UG-135/U)	---	0.30	AL
	04							M3922/53-001 (UG-39/U)	---		COP
	05							M3922/53-001 (UG-39/U)	---		CRES
M3954/22	001			1.15:1	2000		---	M3922/53-001 (UG-39/U)	---	0.2	AL
M3954/22	002	Same as -001 except has resistor, fuse, and thermopile.									
M3954/21	01	10.25	10.50	1.15:1	7000	---	40	(UG-80/U)	---	2	CRES
M3954/13	02	12.40	18.00		500	160	30	M3922/53-006 (UG-1665/U)	---	0.15	AL
	03							M3922/53-005 (UG-419/U)	---		COP
	04							M3922/53-005 (UG-419/U)	---		CRES
CLASS III - NONFINNED AND NONLIQUID COOLED WAVEGUIDE LOADS											
M3954/18	05	2.60	3.95	1.1:1	750	4000	10	M3922/52-010 (UG-1725/U)	---	---	AL
M3954/16	05	5.85	8.20	1.1:1	2	2	---	M3922/55-001 (UG-344/U)	---	---	COP

TABLE II. MIL-D-39030 coaxial dash numbers and characteristics.

Part number	Dash	Frequency range (GHz) Min	Frequency range (GHz) Max	VSWR (max)	Power Average	Power Peak	Finish	Connector	Body material	Dimensions L	Dimensions W	AN nomenclature
					W	kW						
TYPE I - SERIES SMA CONNECTORS												
M39030/3	15	4.4	5.0	1.05:1	0.5	0.05	GLD	SMA-male	CRES	.56	.28	---
M39030/3	09	DC	10.0	1.10:1	1.0	0.10	GLD		CRES	.84	.26	---
M39030/3	10		10.0	2.10:1	2.0	0.20	GLD		CRES	.55	.26	---
M39030/4	01		12.4	1.30:1	10.0	3.0	---		AL	2.0	1.25	---
M39030/4	02		12.4	1.30:1	10.0	3.0	---		CRES	2.0	1.25	---
M39030/3	01		18.0	1.23:1	0.5	0.05	GLD			.73	.28	---
	02						PSVT			.73	.28	---
	03						GLD			.85	.36	---
	04						PSVT			.85	.36	---
	05						GLD	SMA-female		.72	.31	---
	06						PSVT			.72	.31	---
	07						GLD			.80	.38	---

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TABLE II. MIL-D-39030 coaxial dash numbers and characteristics - Continued.

Part number	Dash	Frequency range (GHz) Min	Frequency range (GHz) Max	VSWR (max)	Power Average W	Power Peak kW	Finish	Connector	Body material	Dimensions L	Dimensions W	AN nomenclature
TYPE I - SERIES SMA CONNECTORS												
M39030/3	08	DC	18.0	1.23:1	0.5	0.05	PSVT	SMA-female	CRES	.80	.38	---
↓	11	DC	18.0	1.15:1	1.0	0.10	GLD	SMA-male	↓	.43	.27	---
	12	2.0	19.0	1.30:1	1.0	0.10	↓	↓		.55	.26	---
	13	↓	↓	↓	0.5	0.05				.61	.26	---
	14				1.0	0.10				.55	.26	---
TYPE III - SERIES BNC CONNECTORS												
M39030/7	01	DC	.25	1.10:1	0.5	1.0	GLD	BNC-male	CRES	1.42	.50	---
↓	02			↓	↓	↓	↓	↓	↓	↓	↓	---
	03											---
	04											---
	05			1.15:1	2.0		NP		BRS	1.00		---
	06		↓	1.15:1	2.0	↓	NP	BNC-female	BRS	1.00	↓	---
M39030/1	03		5.0	1.50:1	600	2.5	---	BNC-male	---	---	---	DA-495/U
M39030/1	04	↓	5.0	1.50:1	600	2.5	---	BNC-female	---	---	---	DA-496/U
TYPE IV - SERIES TNC CONNECTORS												
M39030/5	02	.06	.08	1.12:1	0.5	0.6	GLD	TNC-male	CRES	1.29	.50	---
M39030/1	05	DC	5.0	1.50:1	600	2.5	---	TNC-male	---	---	---	DA-497/U
M39030/1	06		5.0	1.50:1	600	2.5	---	TNC-female	---	---	---	DA-498/U
M39030/8	01		6.0	1.30:1	10.0	5.0	ANDZ	TNC-male	AL	2.25	1.39	---
M39030/9	01		8.0	1.40:1	25.0	5.0	ANDZ	↓	AL	4.35	1.39	---
M39030/5	01		10.0	1.40:1	5.0	6.0	GLD		CRES	1.52	0.70	---
↓	03		↓	1.15:1	2.0	2.5	↓			1.30	---	---
	04			1.20:1	5.0	6.0				1.50	.69	---
	05	↓	11.0	1.10:1	3.0	1.0	PSVT	↓		1.33	.50	---
	06	.03	12.4	↓	1.0	1.2	NP			1.32	↓	---
	07	.03	12.4		1.0	1.2	NP	TNC-female	↓	1.20		---
TYPE V - SERIES N CONNECTORS												
M39030/1	01	DC	5.0	1.50:1	600	10.0	---	N-male	---	---	---	DA-494/U
M39030/1	02		5.0	1.50:1	600	10.0	---	N-female	---	---	---	DA-492/U
M39030/10	01		6.0	1.30:1	10.0	5.0	ANDZ	N-male	AL	2.25	1.39	---
M39030/11	01		8.0	1.40:1	50.0	5.0	↓	↓	↓	4.61	1.63	---
↓	02		10.0	1.30:1	40.0	7.5				5.06	1.63	---
	03		10.0	1.30:1	40.0	7.5		N-female		4.91	1.66	---
M39030/6	05		12.4	1.15:1	5.0	2.0		N-male		1.64	.70	---
↓	01	↓	↓	1.20:1	5.0	2.0	GLD	↓	CRES	1.64	.76	---
	02			1.35:1	1.0	1.0	GLD		↓	1.90	.76	---
	03	.03		1.10:1	1.0	0.5	NP			1.51	.38	---

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TABLE II. MIL-D-39030 coaxial dash numbers and characteristics - Continued

Part number	Dash	Frequency range (GHz) Min	Frequency range (GHz) Max	VSWR (max)	Power Average	Power Peak	Finish	Connector	Body material	Dimensions L	Dimensions W	AN nomenclature
					W	kW						
TYPE V - SERIES N CONNECTORS												
M39030/6	04	.03	12.4	1.10:1	1.0	0.5	NP	N-female	CRES	1.48	.38	---
	06	DC	18.0		2.0	1.0	PSVT	N-male		1.28	.81	---
	07				2.0	1.0	PSVT	N-female		1.28	.81	---
M39030/10	02			1.30:1	20.0	5.0	ANDZ		AL	2.28	1.28	---
M39030/11	04	0.7		1.20:1	175	10.0	ANDZ		AL	12.94	2.44	---
TYPE VI - SERIES C CONNECTORS												
M39030/1	09	DC	5.0	1.50:1	600	10.0	---	C-male	---	---	---	---
M39030/1	10		5.0	1.50:1	600	10.0	---	C-female	---	---	---	---
M39030/12	01		12.4	1.30:1	10.0	5.0	GLD	C-male	CRES	1.76	.76	---
TYPE IX - SERIES LC CONNECTORS												
M39030/2	01	DC	3.5	1.30:1	500	50	---	LC/mod	---	---	---	DA-75/U
M39030/2	02		3.5	1.30:1	500	50	---	LC/mod	---	---	---	DA-504/U
M39030/1	07		5.0	1.50:1	600	250	---	LC-male	---	---	---	DA-499/U
M39030/1	08		5.0	1.50:1	600	250	---	LC-female	---	---	---	DA-500/U
TYPE XII - 1-5/8 INCH COAXIAL												
M39030/1	11	DC	5.0	1.50:1	600	250	---	1-5/8 in. coax	---	---	---	DA-503/U

TABLE III. MIL-D-39030 stripline dash numbers and characteristics.

Part number	Dash	Frequency range (GHz) Min	Frequency range (GHz) Max	VSWR (max)	Power Average	Power Peak	Finish	Body material	Dimensions L	Dimensions W
					W	kW				
TYPE XIV - SOCKET CONTACT										
M39030/13	01	DC	12.4	1.25:1	1.0	1.0	GLD	AL	.32	.50
TYPE XIV - BIFURCATED CONTACT										
M39030/15	01	DC	12.0	1.50:1	20.0	0.5	CHEM CVRSN CTD	AL	.281	.500
	02									
	03									
M39030/14	01		18.0	1.40:1	1.0	0.2			.170	
	02									
	03									
M39030/16	01									1.00
	02									
	03									
TYPE XIV - TAB CENTER CONTACT										
M39030/19	01	DC	12.0	1.50:1	20.0	20.0	CHEM CVRSN CTD	AL	.281	.500
M39030/19	02		12.0	1.50:1	20.0	20.0	CHEM CVRSN CTD	AL	.281	
M39030/17	01		12.4	1.30:1	1.0	1.0	GLD	BRS	.165	

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TABLE III. MIL-D-39030 stripline dash numbers and characteristics - Continued

Part number	Dash	Frequency range (GHz) Min	Frequency range (GHz) Max	VSWR (max)	Power Average	Power Peak	Finish	Body material	Dimensions L	Dimensions W
					W	kW				
TYPE XIV - TAB CONTACT										
M39030/21	01	DC	3.0	1.15:1	3.0	0.1	GLD	BRS	---	---
M39030/22	01	↓	4.0	1.25:1	40.0	0.3	↓	CRES	---	---
M39030/20	01	↓	6.0	1.35:1	1.0	0.1	↓	BRS	---	---
M39030/21	02	↓	6.0	1.35:1	3.0	↓	↓	↓	---	---
M39030/20	02	↓	12.0	1.40:1	1.0	↓	↓	↓	---	---
M39030/20	03	↓	↓	↓	1.0	↓	↓	↓	---	---
M39030/21	03	↓	↓	↓	3.0	↓	↓	↓	---	---
M39030/21	04	↓	12.4	1.25:1	3.0	3.0	↓	↓	---	---
M39030/20	05	↓	↓	1.35:1	0.5	0.5	↓	↓	---	---
M39030/20	04	↓	↓	1.35:1	1.0	0.1	↓	↓	---	---
M39030/20	06	↓	18.0	1.25:1	1.0	1.0	NP	↓	---	---
TYPE XIV - PIN CONTACT										
M39030/18	01	4.40	5.00	1.10:1	0.50	0.50	PSVT	CRES	.525	.50
M39030/18	02	7.125	8.40	1.30:1	0.50	0.50	PSVT	CRES	.525	.50

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - MI, AR
Navy - OS, SH
Air Force - 17, 85
DLA - ES

User activities:
Army - AV
Navy - AS, CG, MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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